香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 5-6: Linearization

$$u \to \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \end{pmatrix} \to y$$

use output to calculate parameter back!

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$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$y = 3x$ linear

$$\left.\begin{array}{l} y = x^2 \\ y = \sin x \end{array}\right\} \text{non-linear}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & \sin\theta & 0 \end{bmatrix}$$

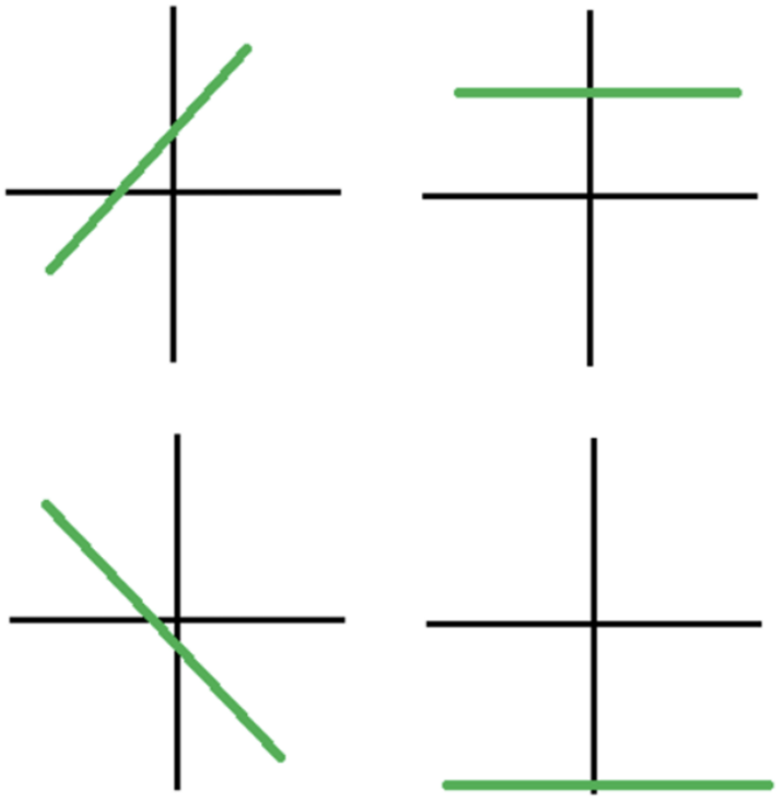
$\uparrow$

?. deal

approximate

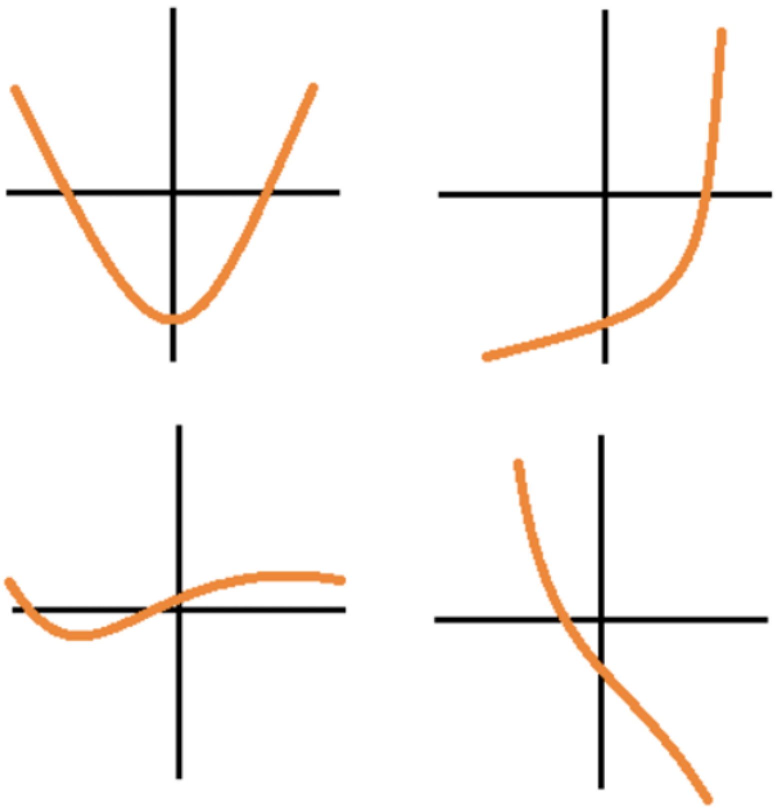
Linear v.s. Nonlinear System

Linear



For example: $y=f(x)=x$

Nonlinear



For example: $y=f(x)=x^2$

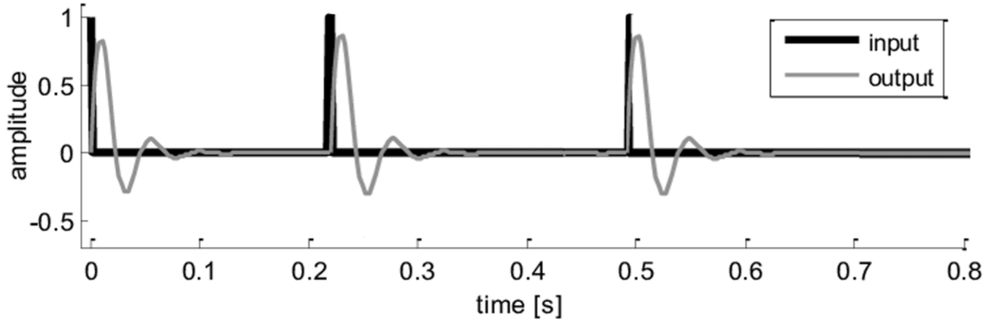
only deal with this

Time-invariant v.s. Time-variant System

LTI!

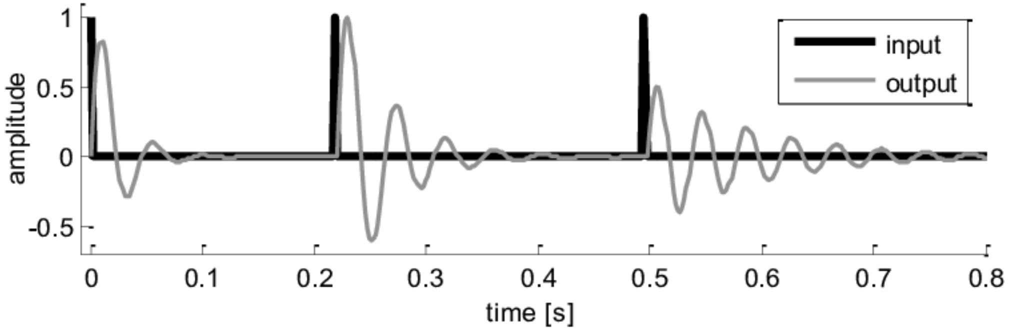
Time-invariant system

car driving



When given the same input to the system at different times, the output is the same.

Time-variant system



When given the same input to the system at different times, the output is different.

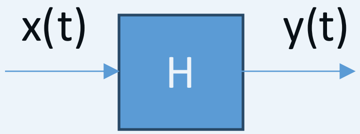
Linear Time-invariant Systems

- **Definition** A system is said to be **linear** if it can be described by linear differential equations, in particular, if the functions $\boldsymbol{f}$ and g in its state space model are linear functions of $\boldsymbol{x}(t)$ and $u(t)$.

- **Definition** A system is said to be **time-invariant** if it can be described by differential equations with constant coefficients, in particular, if the functions $\boldsymbol{f}$ and g in its state space model do not depend on the time t explicitly.

$y_1 = 3x_1 \quad y_2 = 3x_2$

$a_1 \cdot 3x_1 + a_2 \cdot 3x_2 = 3(a_1 y_1 + a_2 y_2)$



Mathematically, a system is linear time-invariant if it satisfies the following conditions for all inputs $x_1(t)$ and $x_2(t)$ and constants a and b:

linear

1. **Superposition:**

   $$H[a \cdot x_1(t) + b \cdot x_2(t)] = a \cdot H[x_1(t)] + b \cdot H[x_2(t)]$$

2. **Scaling:**

   $$H[a \cdot x(t)] = a \cdot H[x(t)]$$

3. **Time-Invariance:**

   $$H[x(t - \tau)] = y(t - \tau)$$

Linear time-invariant (LTI) Systems

- **State space model of a linear time-invariant (LTI) system**

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{b}u(t)$$
$$y(t) = \boldsymbol{c}\boldsymbol{x}(t) + du(t)$$

where $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is an $n \times n$ constant matrix, $\boldsymbol{b} \in \mathbb{R}^{n\times 1}$ and $\boldsymbol{c} \in \mathbb{R}^{1\times n}$ are, respectively, constant column and row vectors, and $d \in \mathbb{R}$ is a constant.

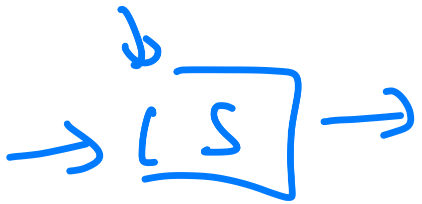
- **LTI systems example: a simple RLC circuit**

$$\begin{bmatrix} \dot{v}_1(t) \\ \dot{v}_2(t) \\ \dot{i}(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1C_1} & 0 & -\frac{1}{C_1} \\ -\frac{g}{C_2} & -\frac{1}{R_2C_2} & \frac{1}{C_2} \\ \frac{1}{L} & -\frac{1}{L} & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \\ i(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1C_1} \\ 0 \\ 0 \end{bmatrix} v_i(t),$$

Constant!

$$v_o(t) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \\ i(t) \end{bmatrix}.$$

Superposition Principle



- **Superposition Principle** Assume that a linear system has **zero initial condition**. If input $u_1(t)$ produces output $y_1(t)$ and input $u_2(t)$ produces output $y_2(t)$, then input $\alpha_1 u_1(t) + \alpha_2 u_2(t)$ produces output $\alpha_1 y_1(t) + \alpha_2 y_2(t)$ for all $\alpha_1, \alpha_2 \in \mathbb{R}$.

  - **Proof** With zero initial condition, if input $u_1(t)$ produces output $y_1(t)$ and input $u_2(t)$ produces output $y_2(t)$, then there are $\boldsymbol{x}_1(t)$ and $\boldsymbol{x}_2(t)$ with $\boldsymbol{x}_1(0) = \mathbf{0}$ and $\boldsymbol{x}_2(0) = \mathbf{0}$ satisfying

$$\begin{aligned} \dot{\boldsymbol{x}}_1(t) &= \boldsymbol{A}\boldsymbol{x}_1(t) + \boldsymbol{b}u_1(t) \\ y_1(t) &= \boldsymbol{c}\boldsymbol{x}_1(t) + du_1(t) \end{aligned}$$

and

$$\begin{aligned} \dot{\boldsymbol{x}}_2(t) &= \boldsymbol{A}\boldsymbol{x}_2(t) + \boldsymbol{b}u_2(t) \\ y_2(t) &= \boldsymbol{c}\boldsymbol{x}_2(t) + du_2(t). \end{aligned}$$

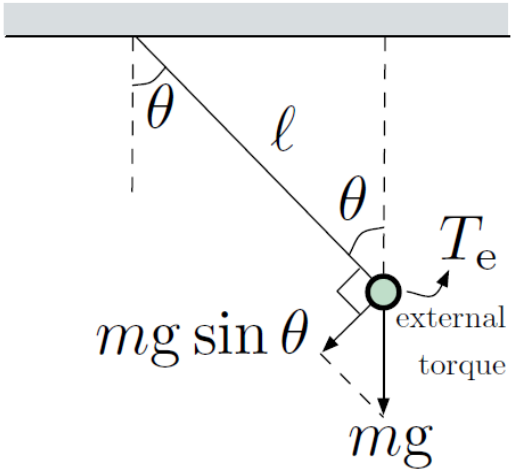
Superposition Principle

- **Superposition Principle** Assume that a linear system has **zero initial condition**. If input $u_1(t)$ produces output $y_1(t)$ and input $u_2(t)$ produces output $y_2(t)$, then input $\alpha_1 u_1(t) + \alpha_2 u_2(t)$ produces output $\alpha_1 y_1(t) + \alpha_2 y_2(t)$ for all $\alpha_1, \alpha_2 \in \mathbb{R}$.
  - **Proof** If we add the two state equations and the two output equations, respectively, and define $u(t) = \alpha_1 u_1(t) + \alpha_2 u_2(t)$, $\boldsymbol{x}(t) = \alpha_1 \boldsymbol{x}_1(t) + \alpha_2 \boldsymbol{x}_2(t)$, and $y(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$, then we obtain $\boldsymbol{x}(0) = \boldsymbol{0}$ and

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{b}u(t)$$
$$y(t) = \boldsymbol{c}\boldsymbol{x}(t) + du(t).$$

$\square$

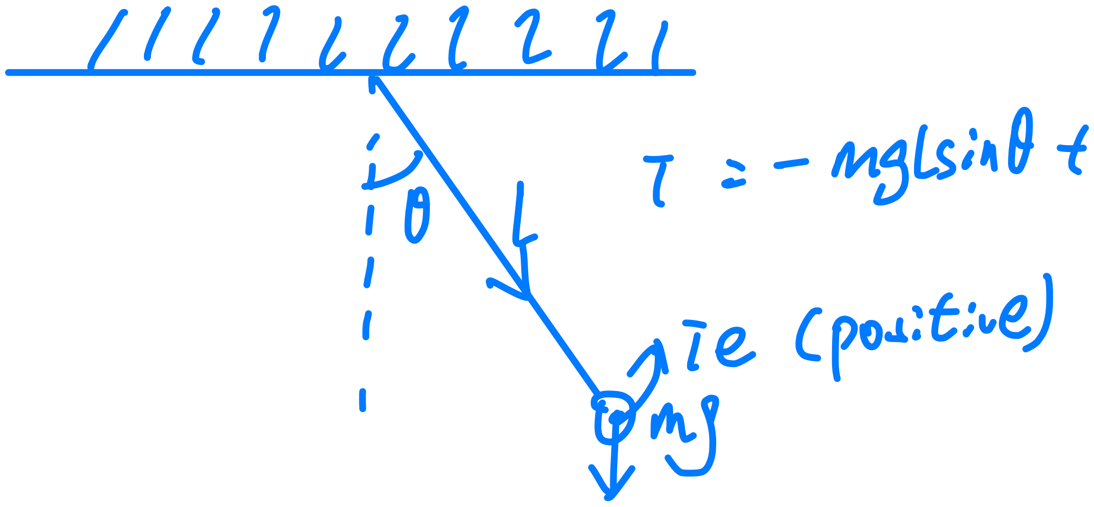
Example 3: Pendulum



$$J\ddot{\theta} = T_e - mg\sin\theta\,\ell$$

$\dot{x} = Ax + Bu$

$y = Cx$



$\tau = -mgl\sin\theta + \tau_e$

$F = ma$

$ma = F_1 - F_2$

$\tau = \Sigma$

Need to derive the direction!

$\tau = J\ddot{\theta}$ rotational $\approx M\ddot{x}$ translation

$I = ml^2$ like $F = ma$

$\Rightarrow ml^2\ddot{\theta} = \tau_e - mg\sin\theta L$

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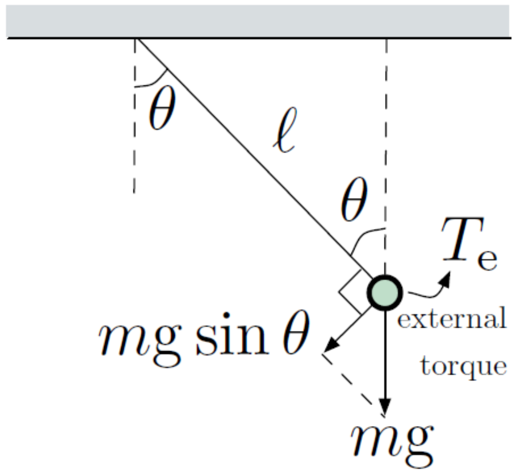
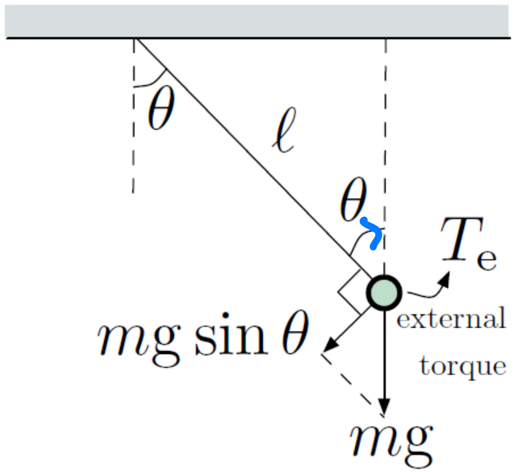


Newton's 2nd law (rotational motion):

Example 3: Pendulum



Newton's 2nd law (rotational motion):

$$\underbrace{T}_{\substack{\text{total}\\\text{torque}}} = \underbrace{J}_{\substack{\text{moment}\\\text{of inertia}}} \underbrace{\alpha}_{\substack{\text{angular}\\\text{acceleration}}}$$

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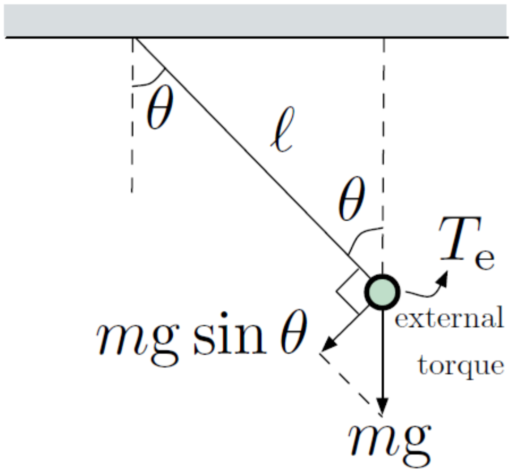

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$$= \text{pendulum torque} + \text{external torque}$$

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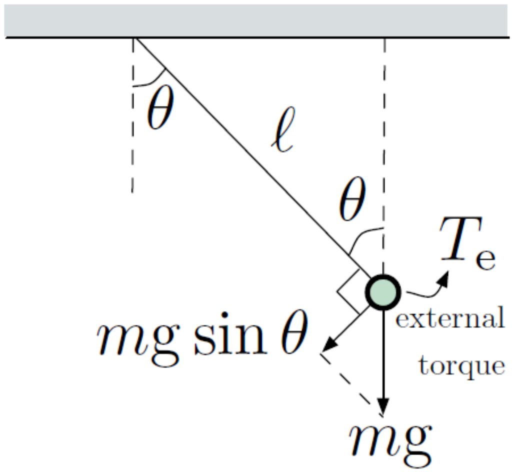

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$$\text{pendulum torque} = \underbrace{-mg\sin\theta}_{\text{force}} \cdot \underbrace{\ell}_{\text{lever arm}}$$

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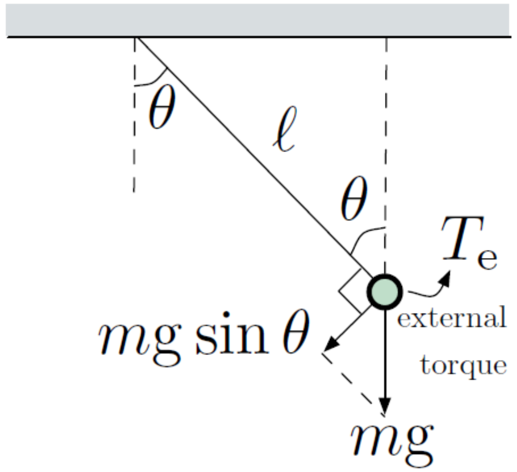

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$$\text{moment of inertia } J = m\ell^2$$

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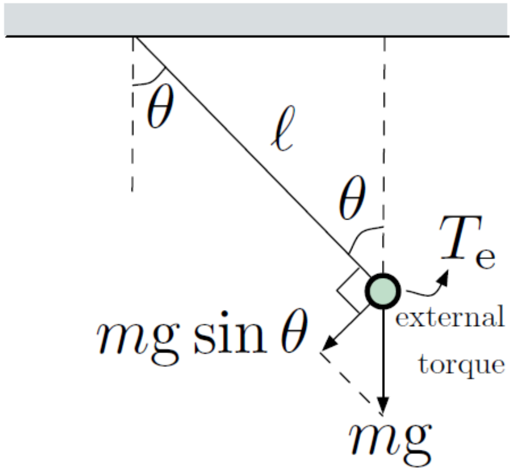

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$$-mg\ell\sin\theta + T_e = m\ell^2\ddot{\theta}$$

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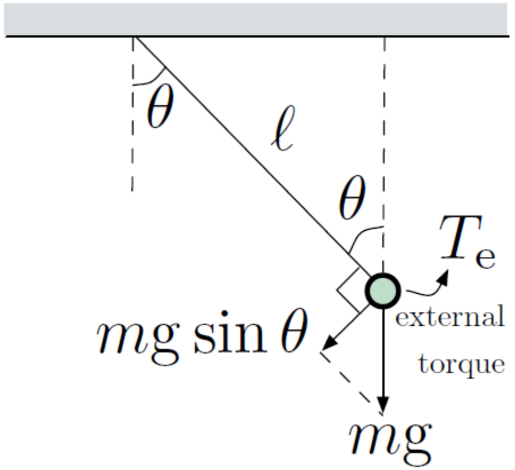

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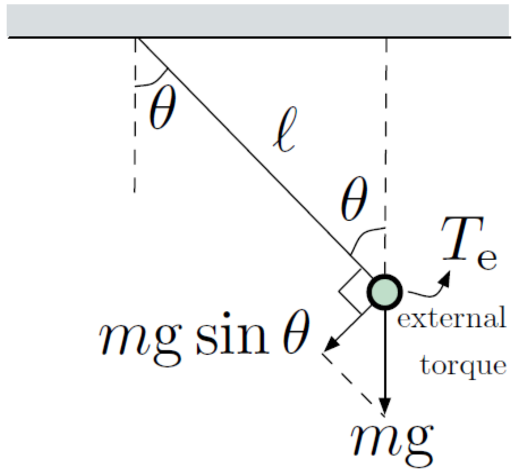

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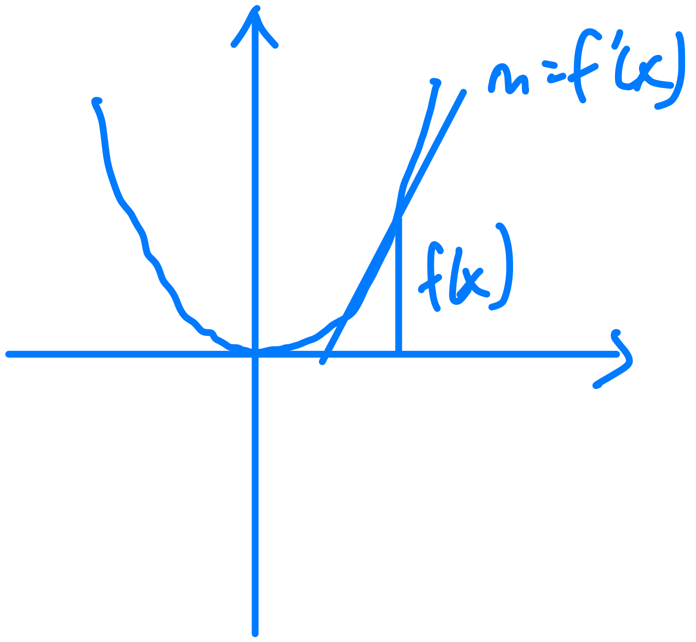
$$-mg\ell\sin\theta + T_e = m\ell^2\ddot{\theta}$$

$$\ddot{\theta} = -\frac{g}{\ell}\sin\theta + \frac{1}{m\ell^2}T_e \qquad \text{(nonlinear equation)}$$

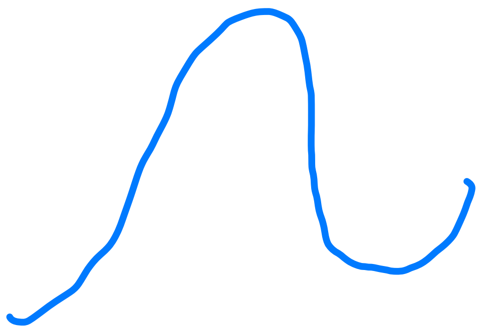
$$\begin{pmatrix} \dot{\theta} \\ \ddot{\theta} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ ? & ? \end{pmatrix} \begin{pmatrix} \theta \\ \dot{\theta} \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{ml^2} \end{pmatrix} T_e$$

don't know how to handle it?



$\sin\theta$

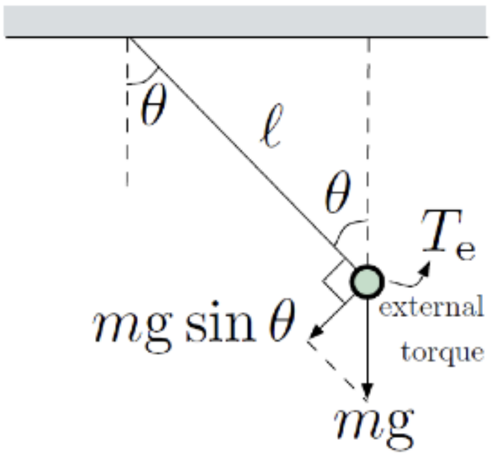
$$f(x) = f(x_0) + f'(x_0)\,\Delta x + \dots$$



$$\dot{x} = f(x_0, u)$$

$\sin\theta \to \theta$ when $\theta \to 0$

Example 3: Pendulum



$$\ddot{\theta} = -\frac{g}{\ell}\sin\theta + \frac{1}{m\ell^2}T_e \qquad \text{(nonlinear equation)}$$

$$\dot{\theta}_1 = \theta_2 = f_1(\theta_1, \theta_2, T_e)$$

θ

$$\dot{\theta}_2 = -\frac{g}{l}\sin\theta + \frac{1}{ml^2}T_e = f_2(\theta_1, \theta_2, T_e)$$

Nonlinear

Define state vector: $\boldsymbol{x} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$

⇩

State space form: $\dot{\boldsymbol{x}} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ ? & ? \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} T_e$

$-\frac{g}{l}$ $\quad 0$

Can we convert it to linear approximately?

Example 3: Pendulum

$$\ddot{\theta} = -\frac{\mathrm{g}}{\ell}\sin\theta + \frac{1}{m\ell^2}T_{\mathrm{e}} \qquad \text{(nonlinear equation)}$$

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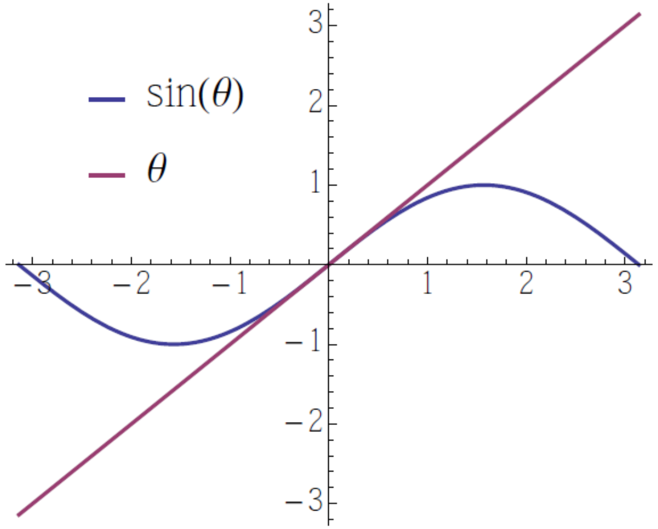

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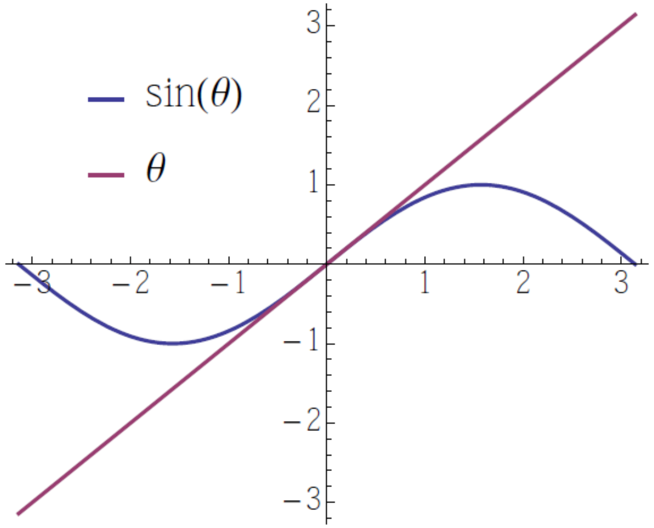

$$\ddot{\theta} = -\frac{\mathrm{g}}{\ell}\theta + \frac{1}{m\ell^2}T_{\mathrm{e}}$$

State-space form: $\theta_1 = \theta,\ \theta_2 = \dot{\theta}$

$$\dot{\theta}_2 = -\frac{\mathrm{g}}{\ell}\theta + \frac{1}{m\ell^2}T_{\mathrm{e}} = -\frac{\mathrm{g}}{\ell}\theta_1 + \frac{1}{m\ell^2}T_{\mathrm{e}}$$

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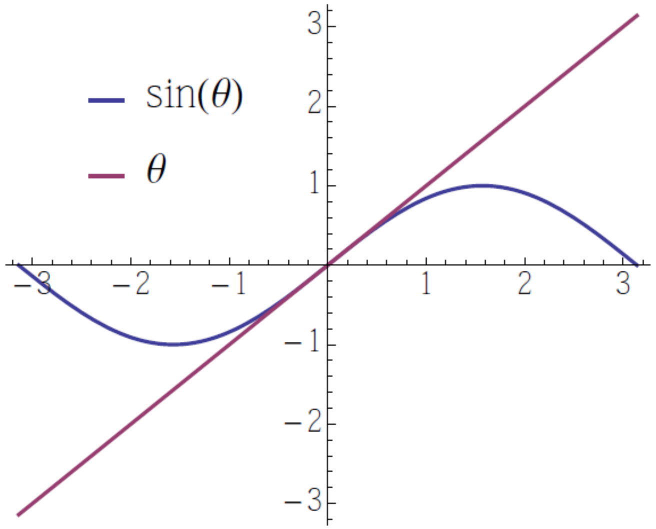

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$$\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\end{pmatrix} = \begin{pmatrix}0 & 1\\ -\frac{\mathrm{g}}{\ell} & 0\end{pmatrix}\begin{pmatrix}\theta_1\\ \theta_2\end{pmatrix} + \begin{pmatrix}0\\ \frac{1}{m\ell^2}\end{pmatrix}T_{\mathrm{e}}$$

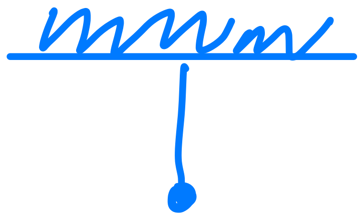
Linearization

Taylor series expansion:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + \dots$$

$$\approx f(x_0) + f'(x_0)(x - x_0) \qquad \text{linear approximation around } x = x_0$$

Control systems are generally *nonlinear*:

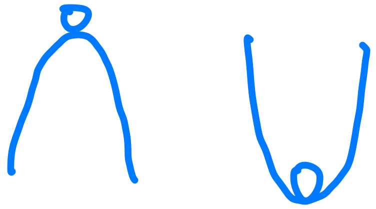


$$\boxed{\dot{x} = f(x, u)} \qquad \text{nonlinear state-space model}$$

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad u = \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix} \quad f = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix}$$

But no need $x = u = 0$

The equilibrium condition is the state we want!



Assume $x = 0, u = 0$ is an *equilibrium point*: $f(0, 0) = 0$

$\dot{x} = 0$ speed not change!

This means that, when the system is at rest and no control is applied, the system does not move.

An equilibrium point is a point at which the rates-of-change for all of the state variables are zero.

Linearization

Linear approx. around $(x, u) = (0, 0)$ to all components of f:

$$\dot{x}_1 = f_1(x, u), \qquad \ldots, \qquad \dot{x}_n = f_n(x, u)$$

For each $i = 1, \ldots, n$,

$$\begin{aligned} f_i(x, u) = \underbrace{f_i(0, 0)}_{=0} &+ \frac{\partial f_i}{\partial x_1}(0, 0)x_1 + \ldots + \frac{\partial f_i}{\partial x_n}(0, 0)x_n \\ &+ \frac{\partial f_i}{\partial u_1}(0, 0)u_1 + \ldots + \frac{\partial f_i}{\partial u_m}(0, 0)u_m \end{aligned}$$

Linearized state-space model:

$$\dot{x} = Ax + Bu, \qquad \text{where } A_{ij} = \left.\frac{\partial f_i}{\partial x_j}\right|_{\substack{x=0\\u=0}}, \quad B_{ik} = \left.\frac{\partial f_i}{\partial u_k}\right|_{\substack{x=0\\u=0}}$$

Important: since we have ignored the higher-order terms, this linear system is only an *approximation* that holds only for *small deviations* from equilibrium.

Example 3: Pendulum, Revisited

Original nonlinear state-space model:

$$\dot{\theta}_1 = f_1(\theta_1, \theta_2, T_e) = \theta_2 \qquad \text{— already linear}$$

$$\dot{\theta}_2 = f_2(\theta_1, \theta_2, T_e) = -\frac{g}{\ell}\sin\theta_1 + \frac{1}{m\ell^2}T_e$$

$$\dot{x} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ ? & ? \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} T_e$$

$$\left.\frac{\partial f_2}{\partial \theta_1}\right|_0 \qquad \left.\frac{\partial f_2}{\partial \theta_2}\right|_0 \qquad \left.\frac{\partial f_2}{\partial T_e}\right|_0$$

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Linear approx. of f_2 around equilibrium $(\theta_1, \theta_2, T_\mathrm{e}) = (0,0,0)$:

$$\frac{\partial f_2}{\partial \theta_1} = -\frac{\mathrm{g}}{\ell}\cos\theta_1 \qquad \frac{\partial f_2}{\partial \theta_2} = 0 \qquad \frac{\partial f_2}{\partial T_\mathrm{e}} = \frac{1}{m\ell^2}$$

$$\left.\frac{\partial f_2}{\partial \theta_1}\right|_0 = -\frac{\mathrm{g}}{\ell} \qquad \left.\frac{\partial f_2}{\partial \theta_2}\right|_0 = 0 \qquad \left.\frac{\partial f_2}{\partial T_\mathrm{e}}\right|_0 = \frac{1}{m\ell^2}$$

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$$\dot{\theta}_1 = f_1(\theta_1, \theta_2, T_e) = \theta_2 \qquad \text{— already linear}$$

$$\dot{\theta}_2 = f_2(\theta_1, \theta_2, T_e) = -\frac{g}{\ell} \sin\theta_1 + \frac{1}{m\ell^2} T_e$$

Linear approx. of f_2 around equilibrium $(\theta_1, \theta_2, T_e) = (0, 0, 0)$:

$$\frac{\partial f_2}{\partial \theta_1} = -\frac{g}{\ell} \cos\theta_1 \qquad \frac{\partial f_2}{\partial \theta_2} = 0 \qquad \frac{\partial f_2}{\partial T_e} = \frac{1}{m\ell^2}$$

$$\left.\frac{\partial f_2}{\partial \theta_1}\right|_0 = -\frac{g}{\ell} \qquad \left.\frac{\partial f_2}{\partial \theta_2}\right|_0 = 0 \qquad \left.\frac{\partial f_2}{\partial T_e}\right|_0 = \frac{1}{m\ell^2}$$

Linearized state-space model of the pendulum:

$$\dot{\theta}_1 = \theta_2$$

$$\dot{\theta}_2 = -\frac{g}{\ell}\theta_1 + \frac{1}{m\ell^2} T_e \qquad \text{valid for } \textit{small} \text{ deviations from equ.}$$

General Linearization Procedure

shows the coordinate shift

- Start from nonlinear state-space model

$$\dot{x} = f(x, u)$$

- Find equilibrium point (x_0, u_0) such that $f(x_0, u_0) = 0$

Note: different systems may have different equilibria, not necessarily $(0, 0)$, so we need to shift variables:

$$\underline{x} = x - x_0 \qquad \underline{u} = u - u_0$$
$$\underline{f}(\underline{x}, \underline{u}) = f(\underline{x} + x_0, \underline{u} + u_0) = f(x, u)$$

Note that the transformation is *invertible:*

$$x = \underline{x} + x_0, \qquad u = \underline{u} + u_0$$

General Linearization Procedure

- Pass to shifted variables $\underline{x} = x - x_0$, $\underline{u} = u - u_0$

$$\begin{aligned} \dot{\underline{x}} &= \dot{x} && (x_0 \text{ does not depend on } t) \\ &= f(x, u) \\ &= \underline{f}(\underline{x}, \underline{u}) \end{aligned}$$

— equivalent to original system

- The transformed system is in equilibrium at $(0, 0)$:

$$\underline{f}(0, 0) = f(x_0, u_0) = 0$$

- Now linearize:

$$\dot{\underline{x}} = A\underline{x} + B\underline{u}, \qquad \text{where } A_{ij} = \left.\frac{\partial f_i}{\partial x_j}\right|_{\boxed{\substack{x=x_0 \\ u=u_0}}}, \quad B_{ik} = \left.\frac{\partial f_i}{\partial u_k}\right|_{\substack{x=x_0 \\ u=u_0}}$$

Linearization - discussion

- Why do we require that $f(x_0, u_0) = 0$ in equilibrium?
- This requires some thought. Indeed, we may talk about a *linear approximation* of any smooth function f at any point x_0:

  $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$ — $f(x_0)$ does not have to be 0

- The key is that we want to approximate a given nonlinear system $\dot{x} = f(x, u)$ by a *linear* system $\dot{x} = Ax + Bu$ (may have to shift coordinates: $x \mapsto x - x_0$, $u \mapsto u - u_0$)

  Any linear system *must* have an equilibrium point at $(x, u) = (0, 0)$:

  $$f(x, u) = Ax + Bu \qquad f(0, 0) = A0 + B0 = 0.$$

Linearization - Summary

Assume that a system is described by a state space model, *f* and *g* are continuously differentiable functions:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}[\boldsymbol{x}(t), u(t)]$$
$$y(t) = g[\boldsymbol{x}(t), u(t)]$$

$$\boxed{\begin{aligned}\dot{\boldsymbol{x}}(t) &= \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{b}u(t)\\ y(t) &= \boldsymbol{c}\boldsymbol{x}(t) + du(t)\end{aligned}}$$

state will not change can only observed limited things!

The operating point of the system is $(u_0, \boldsymbol{x}_0, y_0) \in \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}$ $\dot{x} = 0$

$$0 = \boldsymbol{f}(\boldsymbol{x}_0, u_0)$$
$$y_0 = g(\boldsymbol{x}_0, u_0).$$

$$\boxed{\begin{aligned}0 &= Ax_0 + bu_0\\ y_0 &= \mathbf{c}x_0 + du_0\end{aligned}}$$

based on output!

we can approximate the original system by the following linear system:

observe the output!

$$\tilde{u}(t) = u(t) - u_0, \quad \tilde{\boldsymbol{x}}(t) = \boldsymbol{x}(t) - \boldsymbol{x}_0, \quad \tilde{y}(t) = y(t) - y_0.$$

$$\dot{\tilde{\boldsymbol{x}}}(t) = \boldsymbol{A}\tilde{\boldsymbol{x}}(t) + \boldsymbol{b}\tilde{u}(t)$$
$$\tilde{y}(t) = \boldsymbol{c}\tilde{\boldsymbol{x}}(t) + d\tilde{u}(t)$$

$$\boxed{\boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{\boxed{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}}}, \quad \boldsymbol{b} = \left.\frac{\partial \boldsymbol{f}}{\partial u}\right|_{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}}, \quad \boldsymbol{c} = \left.\frac{\partial g}{\partial \boldsymbol{x}}\right|_{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}}, \quad d = \left.\frac{\partial g}{\partial u}\right|_{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}}.}$$

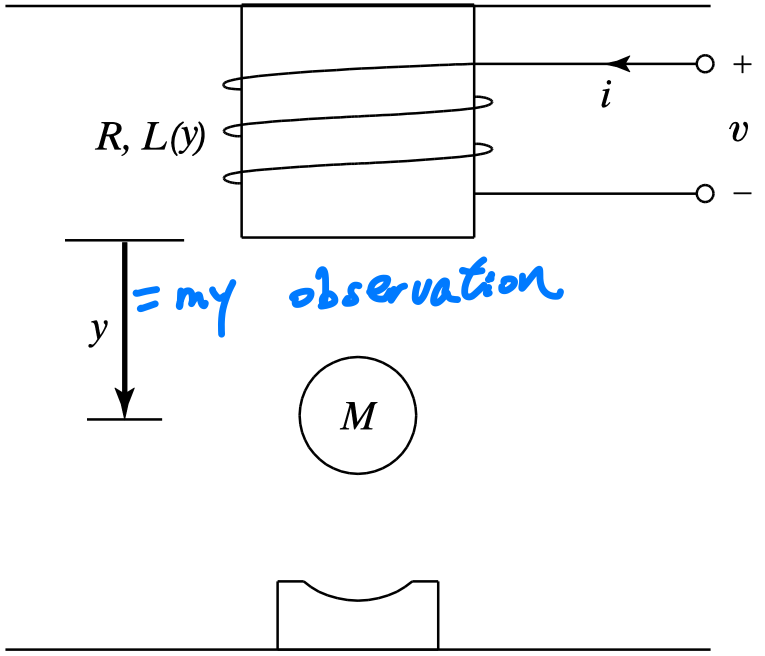
This linear system is called a **linearized system** of the original nonlinear system.

Electromechanical Systems

- **Magnetic levitation systems**

  - The **input** voltage $v(t)$ and **output** position $y(t)$
  - **Intermediate variables:** coil current $i(t)$,
    ball position $y(t)$

    Also can $\dot{y}$
  - **Physical laws:** Kirchhoff's voltage law (KVL),
    Newton's second law



input = $v(t)$

output = y (distance)

$\downarrow$

x: i, y, $\dot{y}$

Electromechanical Systems

- **Magnetic levitation systems**
- The approximate **inductance** function $\boxed{L(y)}$:

Based on Empirical studies $$L(y) = L_0 + \frac{rL_1}{y}$$

where r is the radius of the ball, and L_0 and L_1 are constants.

- KVL applied to the coil:

$$Ri(t) + \frac{dL(y(t))i(t)}{dt} = v(t).$$

- Break the derivative of the product:

$$Ri(t) - \frac{rL_1 i(t)}{y^2(t)}\frac{dy(t)}{dt} + L(y(t))\frac{di(t)}{dt} = v(t).$$

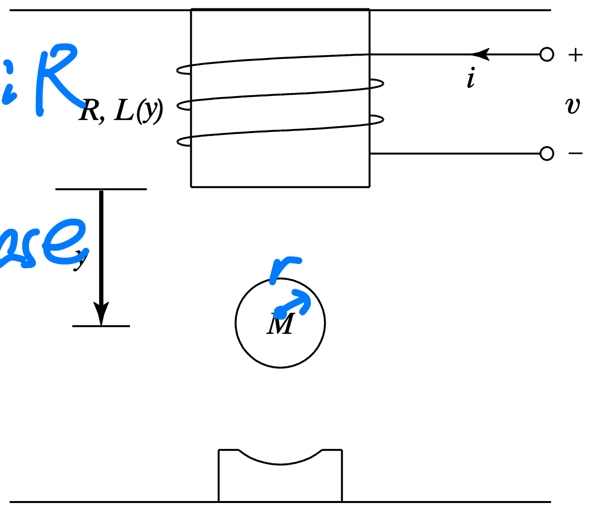
$v = L\frac{di}{dt}$

$V = \frac{L\,di}{dt} + iR$

L is not constant

$V = \frac{dLi}{dt} + iR$

general case



RI

$\frac{dLi}{dt}$

V

$$\frac{d(Li)}{dt} = i\frac{dL}{dt} + L\frac{di}{dt}$$

$$= i\frac{d(L_0 + rL_1 y^{-1})}{dt} + L\frac{di}{dt}$$

$$= i\cancel{\frac{dL_0}{dt}} + rL_1 i(-y^{-2})\dot{y} + L\frac{di}{dt}$$

$$0$$

Electromechanical Systems

- **Magnetic levitation systems**

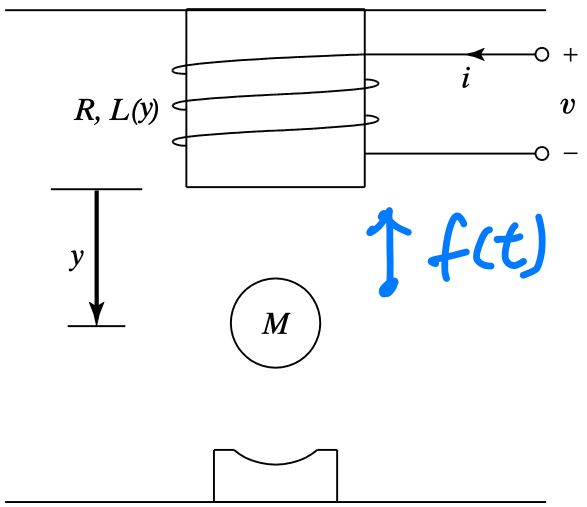
- The **lifting force** with the coil current i and position y:

$$f(t) = \frac{rL_1}{2} \frac{i^2(t)}{y^2(t)}.$$

(see attachment)

- Newton's second law applied to the ball:

$$M \frac{d^2 y(t)}{dt^2} = -\frac{rL_1}{2} \frac{i^2(t)}{y^2(t)} + Mg.$$



State Space Model

- **Example   State space model of MagLev**

The magnetic levitation system has the differential equation model

$$Ri(t) - \frac{rL_1 i(t)}{y^2(t)}\frac{dy(t)}{dt} + L(y(t))\frac{di(t)}{dt} = v(t)$$

$$M\frac{d^2y(t)}{dt^2} = -\frac{rL_1}{2}\frac{i^2(t)}{y^2(t)} + Mg.$$

How to select state variables?

State Space Model

▶ **Example State space model of MagLev**

Choose $x_1(t) = i(t), x_2(t) = y(t), x_3(t) = \dot{y}(t)$ and $u(t) = v(t)$:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} \dfrac{u(t) - Rx_1(t) + rL_1 x_1(t)x_3(t)/x_2^2(t)}{L(x_2(t))} \\ x_3(t) \\ -\dfrac{rL_1}{2M}\dfrac{x_1^2(t)}{x_2^2(t)} + g \end{bmatrix}$$

$$y(t) = x_2(t).$$

Linearization

▶ **Example Linearized magnetic levitation system**

The MagLev is a **nonlinear system** described by state space equation

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} \dfrac{u(t) - Rx_1(t) + rL_1 x_1(t)x_3(t)/x_2^2(t)}{L(x_2(t))} \\ x_3(t) \\ -\dfrac{rL_1}{2M}\dfrac{x_1^2(t)}{x_2^2(t)} + g \end{bmatrix}$$

$$y(t) = x_2(t).$$

– **Task**: suspend the steel ball at **desired position**.

– Hence we need to linearize MagLev around **operating point** (based on desired position $y(t) = y_0$).

Linearization

▶ **Example Linearized magnetic levitation system**

- **Operating Point** Solve equations

$$0 = \frac{u_0 - Rx_{10} + rL_1 x_{10}x_{30}/x_{20}^2}{L(x_{20})}$$

$$0 = x_{30}$$

$$0 = -\frac{rL_1}{2M}\frac{x_{10}^2}{x_{20}^2} + g$$

$$y_0 = x_{20}.$$

This gives

$$u_0 = Ry_0\sqrt{\frac{2Mg}{rL_1}}, \quad \begin{bmatrix} x_{10} \\ x_{20} \\ x_{30} \end{bmatrix} = \begin{bmatrix} y_0\sqrt{\frac{2Mg}{rL_1}} \\ y_0 \\ 0 \end{bmatrix}, \quad y_0 = y_0.$$

Linearization

- **Example Linearized magnetic levitation system**
  - **Linearization Process** Denote the deviations of the input, state, and output variables from the operating point by

$$\tilde{u}(t) = u(t) - u_0 = u(t) - Ry_0\sqrt{\frac{2Mg}{rL_1}}$$

$$\tilde{\boldsymbol{x}}(t) = \boldsymbol{x}(t) - \boldsymbol{x}_0 = \begin{bmatrix} x_1(t) - y_0\sqrt{\dfrac{2Mg}{rL_1}} \\ x_2(t) - y_0 \\ x_3(t) \end{bmatrix}$$

$$\tilde{y}(t) = y(t) - y_0.$$

Linearization

- **Example Linearized magnetic levitation system**
  - **Linearization Process** The **linearized model** of the deviation variables is

$$\dot{\tilde{\boldsymbol{x}}}(t) = \boldsymbol{A}\tilde{\boldsymbol{x}}(t) + \boldsymbol{b}\tilde{u}(t)$$
$$\tilde{y}(t) = \boldsymbol{c}\tilde{\boldsymbol{x}}(t) + d\tilde{u}(t)$$

where

$$\boldsymbol{A} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\bigg|_{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}} = \begin{bmatrix} -\dfrac{R}{L(y_0)} & 0 & \dfrac{\sqrt{2rL_1Mg}}{y_0L(y_0)} \\ 0 & 0 & 1 \\ -\dfrac{1}{y_0}\sqrt{\dfrac{2rL_1g}{M}} & \dfrac{2g}{y_0} & 0 \end{bmatrix}, \quad \boldsymbol{b} = \frac{\partial \boldsymbol{f}}{\partial u}\bigg|_{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}} = \begin{bmatrix} \dfrac{1}{L(y_0)} \\ 0 \\ 0 \end{bmatrix},$$

$$\boldsymbol{c} = \frac{\partial g}{\partial \boldsymbol{x}}\bigg|_{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \qquad d = \frac{\partial g}{\partial u}\bigg|_{\substack{\boldsymbol{x}=\boldsymbol{x}_0\\ u=u_0}} = 0.$$

Review & Exercise

- What is linear and nonlinear system?
- What is time-invariant and time-variant system?
- What the definition of LTI system? What kind of properties does it have?
- The state-space model of LTI system?
- What’s superposition principle?
- What’s the equilibrium point?
- How to convert a nonlinear system to a linear system?
- Please review the relevant mathematic skills and do some exercises by yourself.

- In class exercise:

In-class Exercise

Consider the mechanical system shown in the following figure. Here $F(t)$ is an external force applied to the mass M, $x(t)$ is the displacement of the mass with respect to the position when the spring is relaxed. The spring force is given by $f_{sp}(t) = kx(t) + \alpha x^3(t)$.

1. Write a state space description of the system.

2. Linearize the system around the operating point with $u_0 = 0$.

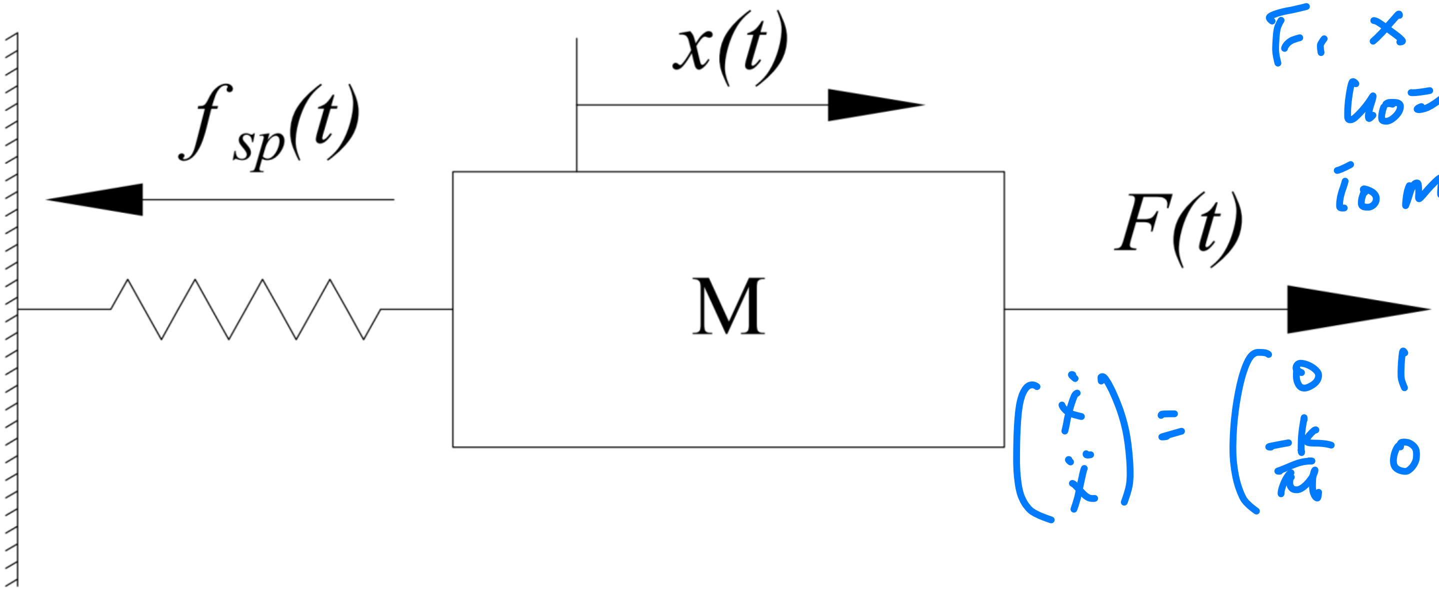
$M\ddot{x} = F - kx + \alpha x^3$

$\ddot{x} = \frac{F}{M} - \frac{k}{M}x + \frac{\alpha}{M}x^3$

F, x

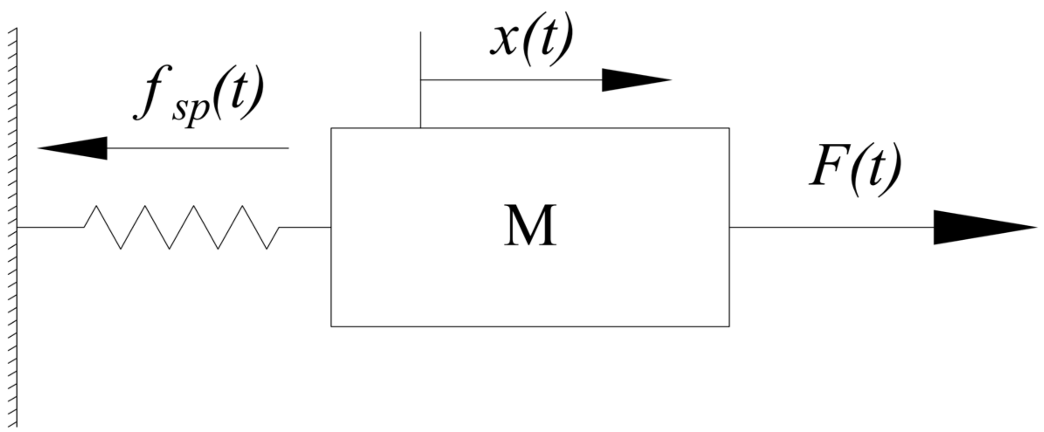
$u_0 = 0$, $x = 0$, $\dot{x} = 0$

to make $\ddot{x} = 0$



$$\begin{pmatrix} \dot{x} \\ \ddot{x} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{M} & 0 \end{pmatrix} \begin{pmatrix} x \\ \dot{x} \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{M} \end{pmatrix} F$$

In-class Exercise



- **Mechanical system**
  - Input: $F(t)$;Output: $x(t)$
  - According to Newton's second law, the differential equation model is

$$m\ddot{x}(t) + \alpha x^3(t) + kx(t) = F(t)$$

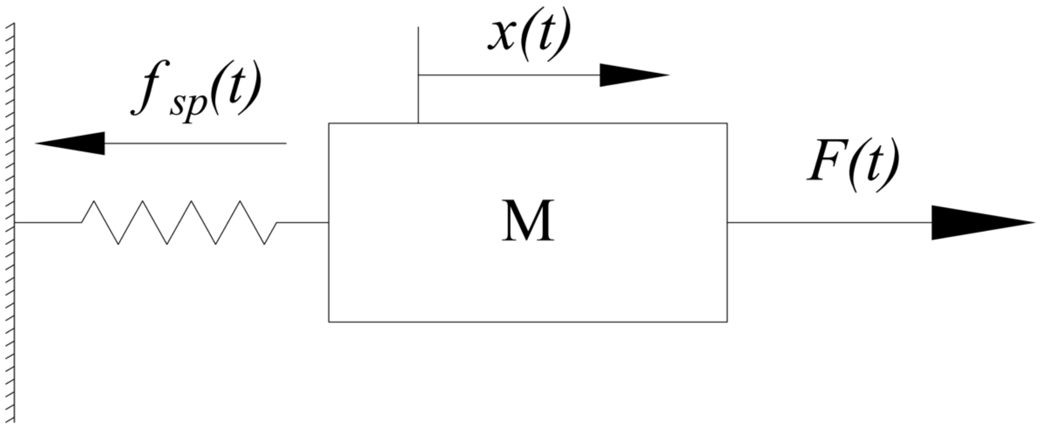
  - Choose $x_1(t) = y(t) = x(t), x_2(t) = \dot{x}(t)$, and $u(t) = F(t)$, the state space model is

$$\begin{bmatrix} \dot{x_1}(t) \\ \dot{x_2}(t) \end{bmatrix} = \begin{bmatrix} x_2(t) \\ -\frac{\alpha}{m}{x_1}^3(t) - \frac{k}{m}x_1(t) + \frac{F}{m} \end{bmatrix}$$

$$y = [1 \;\; 0]\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

In-class Exercise

- **Linearize the mechanical system**



Operation point

Sovel equations:

$$\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix} x_{20} \\ -\frac{\alpha}{m}x_{10}{}^3 - \frac{k}{m}x_{10} + \frac{u_0}{m} \end{bmatrix}$$

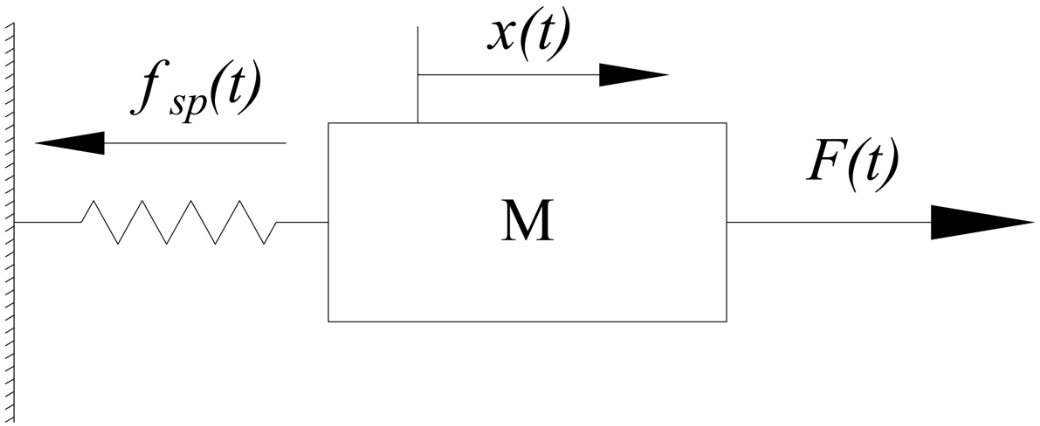
$$y_0 = x_{10}$$

This gives:

$$\begin{bmatrix}x_{10}\\x_{20}\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}, y_0 = 0$$

In-class Exercise

- **Linearization process**



Linearize the system at $f(x_0, u_0) = f(0,0) = 0$

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}$$

$\Rightarrow$

The linearized system model is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

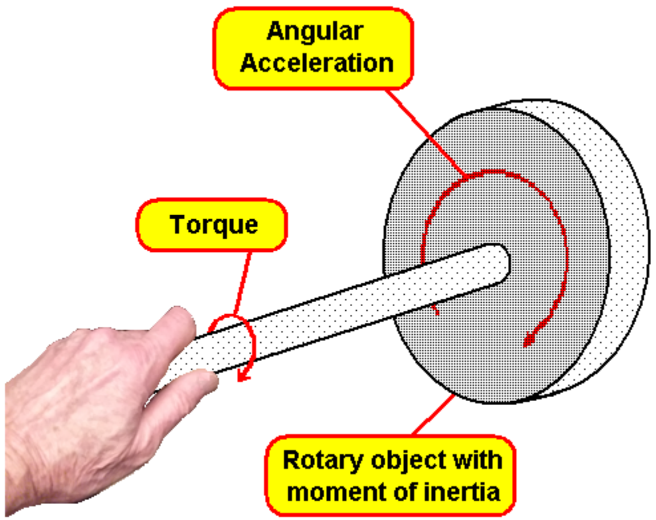
Appendix

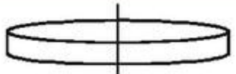
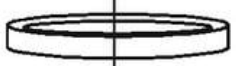
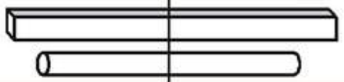
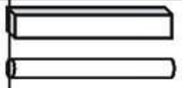
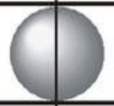
Moment of Inertia

When a body is free to rotate around an axis, torque must be applied to change its angular momentum. The amount of torque needed to cause any given angular acceleration (the rate of change in angular velocity) is proportional to the moment of inertia of the body.

The moment of inertia is defined as the product of mass of section and the square of the distance between the reference axis and the centroid of the section.

$$I_P = \sum_i m_i r_i^2$$



Object	*Drawing*	*Moment of Inertia*
Disk (rotated about center)		$\frac{1}{2}MR^2$
Ring (rotated about center)		MR^2
Rod or plank (rotated about center)		$\frac{1}{12}ML^2$
Rod or plank (rotated about end)		$\frac{1}{3}ML^2$
Sphere		$\frac{2}{5}MR^2$
Satellite		MR^2

https://en.wikipedia.org/wiki/Moment_of_inertia

Taylor Series

If the function *f(x)* and its first *n+1* derivate are continuous on an interval containing x_i and x_{i+1}, then the value of the function at x_{i+1} can be given by:

$$f(x_{i+1}) = f(x_i) + f^{'}(x_i)h + \frac{f^{''}(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \ldots$$

$$+ \frac{f^{(n)}(x_i)}{n!}h^n + R_n \qquad \text{(Eq 4.1)}$$

$$\textit{where}\text{ the step size} \quad \mathrm{h} = x_{i+1} - x_i$$

Remainder: $$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}h^{n+1}$$

ξ is a number lies between x_i and x_{i+1}

Taylor Series

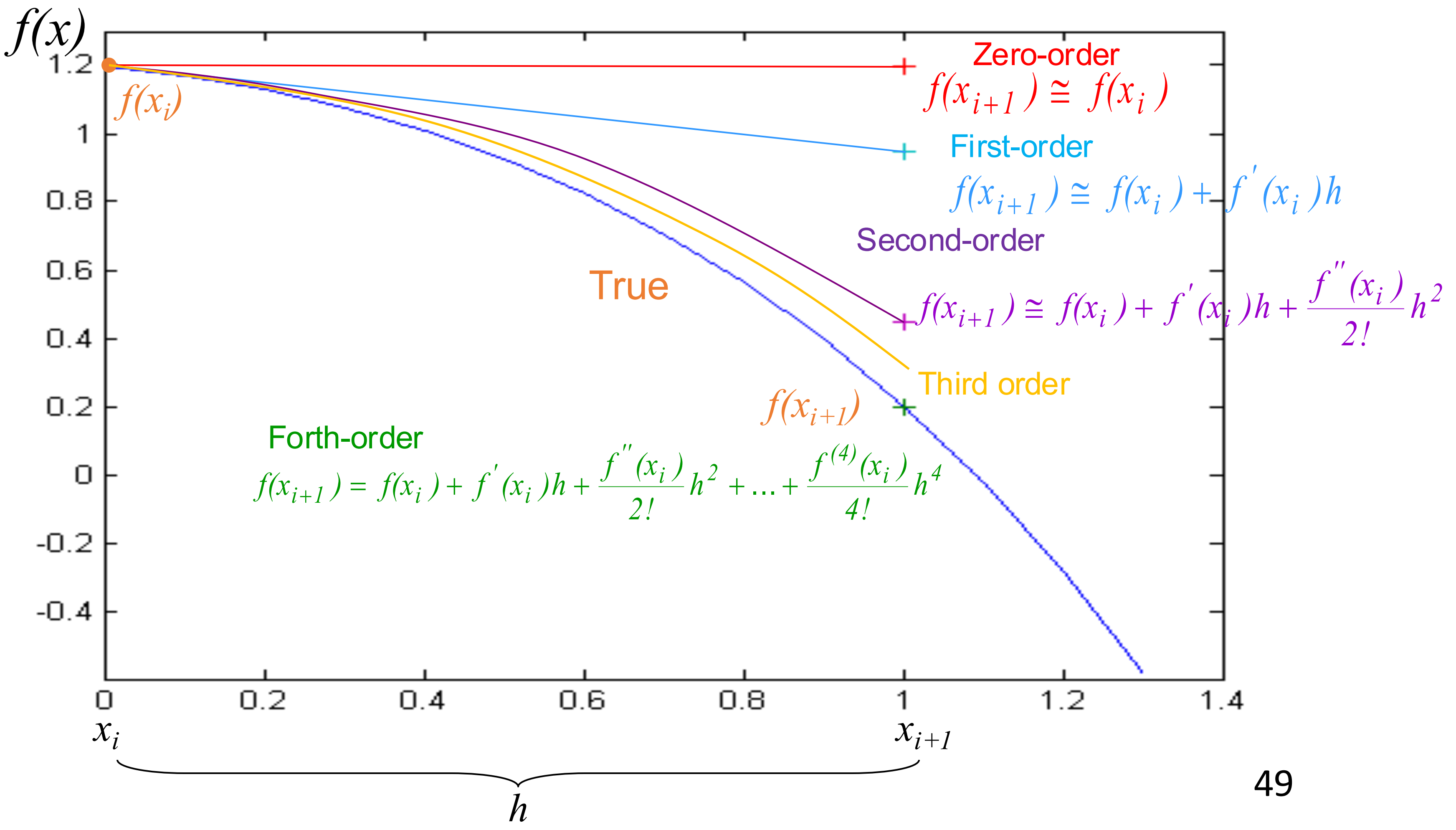
Any smooth function can be approximated as a polynomial and that the Taylor series provides a means to express this idea mathematically.

In general, the n^{th}-order Taylor series expansion can represent exactly for an n^{th}-order polynomial.

Each additional term will contribute some improvement to the approximation.

Luckily in many cases, the inclusion of only a few terms will result in an approximation that is close enough to the true value for practical purposes.

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$



Taylor Series

Taylor expansion in 2D

The Taylor expansion in two dimensions, also known as the Taylor series in two variables, is an extension of the Taylor series to functions of two variables. Let's consider a function *f*(*x*,*y*) and its Taylor expansion around a point (*a*,*b*):

$$f(x,y) = f(a,b) + \left.\frac{\partial f}{\partial x}\right|_{(a,b)}(x-a) + \left.\frac{\partial f}{\partial y}\right|_{(a,b)}(y-b) +$$
$$\frac{1}{2!}\left(\left.\frac{\partial^2 f}{\partial x^2}\right|_{(a,b)}(x-a)^2 + 2\left.\frac{\partial^2 f}{\partial x \partial y}\right|_{(a,b)}(x-a)(y-b) + \left.\frac{\partial^2 f}{\partial y^2}\right|_{(a,b)}(y-b)^2\right.$$
$$\ldots$$

- $f(a,b)$ is the value of the function at the point (a,b).
- $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are the partial derivatives of f with respect to x and y at the point (a,b).
- $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, and $\frac{\partial^2 f}{\partial x \partial y}$ are the second-order partial derivatives of f at the point (a,b).
- The terms with higher-order derivatives follow a similar pattern.

Linearization

The first-order Tylor expansion around the equilibrium point:

nD case!

In the n-dimensional case, let's consider a function $f(\mathbf{X})$ of n variables, where $\mathbf{X}=[x_1,x_2,\ldots,x_n]$ is the vector of variables. At the equilibrium point, i.e., $\mathbf{X}=0$, and $f(\mathbf{X})=0$. The first-order Taylor expansion around the origin $\mathbf{0}$ is given by: given by us! not necessary to be 0!

$$f(\mathbf{X}) \approx \left.\frac{\partial f}{\partial x_1}\right|_{\mathbf{0}} x_1 + \left.\frac{\partial f}{\partial x_2}\right|_{\mathbf{0}} x_2 + \ldots + \left.\frac{\partial f}{\partial x_n}\right|_{\mathbf{0}} x_n$$

$$f_2(x) \approx \left.\frac{\partial f_2}{\partial x_1}\right|_0 x_1 + \frac{\partial f_2}{\partial x_2} x_2$$

- $\left.\frac{\partial f}{\partial x_i}\right|_{\mathbf{0}}$ is the partial derivative of f with respect to x_i evaluated at $\mathbf{0}$.

e.g, if the variables are (x_1, x_2, u):

For $\dot{x} = f(x_1, x_2, u)$ around the equilibrium point

$$f_1(x_1,x_2,u) \approx \left.\frac{\partial f_1}{\partial x_1}\right|_{(0,0,0)} x_1 + \left.\frac{\partial f_1}{\partial x_2}\right|_{(0,0,0)} x_2 + \left.\frac{\partial f_1}{\partial u}\right|_{(0,0,0)} u$$

$$f_2(x_1,x_2,u) \approx \left.\frac{\partial f_2}{\partial x_1}\right|_{(0,0,0)} x_1 + \left.\frac{\partial f_2}{\partial x_2}\right|_{(0,0,0)} x_2 + \left.\frac{\partial f_2}{\partial u}\right|_{(0,0,0)} u$$

differential equation form

$\Leftrightarrow$

In the state space form $\dot{x} = Ax + Bu$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}}_{B} u$$

state-space form

Linearization

How to handle the other cases?

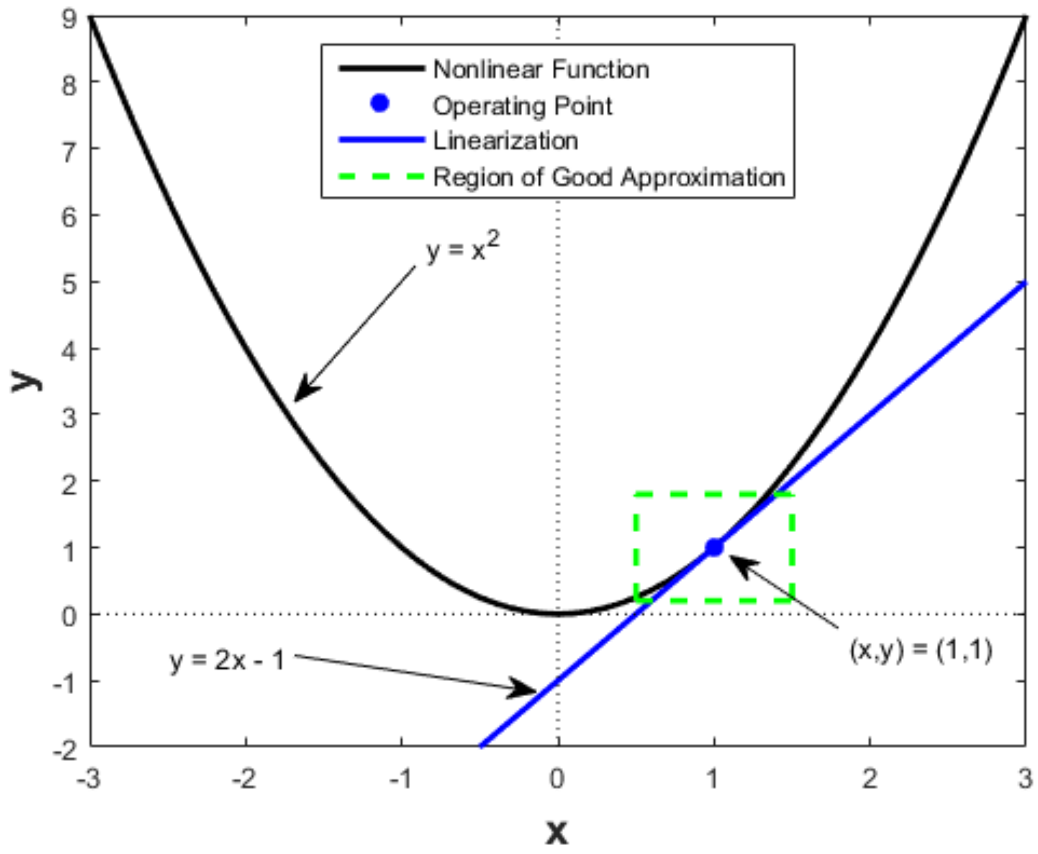
e.g., $y=x^2$, at the point (1, 1)

Ans: $f(x) = f(1)+f'(1)(x-1)$

$= 1+2(x-1)$

$= 2x-1$

Note: $(x^2)'|_{x=1}=2x|_{x=1}=2$



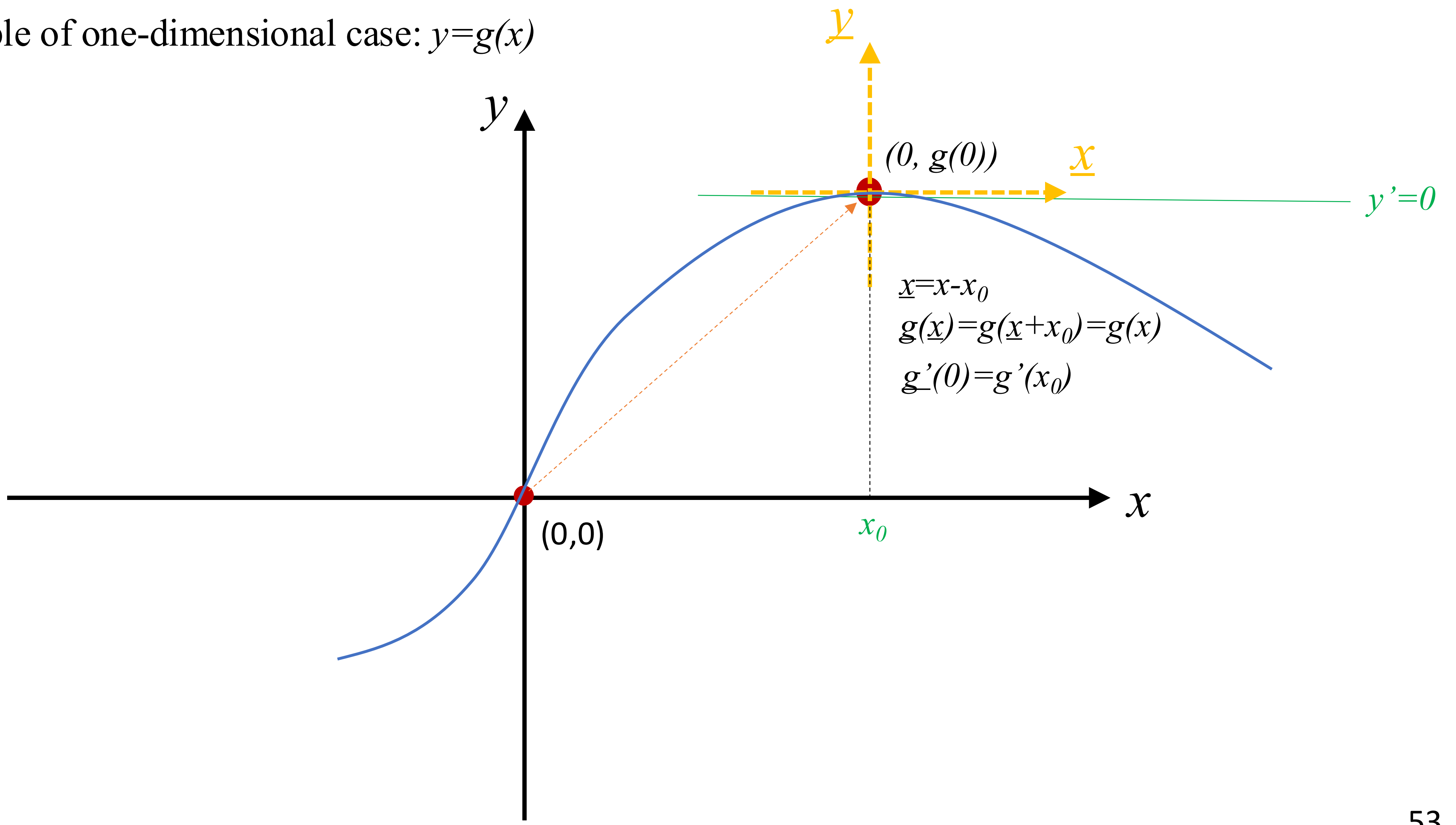
Taylor series expansion:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2 + \ldots$$

$$\approx f(x_0) + f'(x_0)(x - x_0) \qquad \text{linear approximation around } x = x_0$$

We may consider the general linearization problem as a coordinate shift

An example of one-dimensional case: $y=g(x)$



Modeling of the lifting force

- **Magnetic levitation systems**

The total electrical power used to sustain the magnetic field and to move the steal ball is

$$i(t)\frac{dL(y(t))i(t)}{dt} = i^2(t)\frac{dL(y(t))}{dt} + i(t)L(y(t))\frac{di(t)}{dt} \ .$$

The rate of change of the magnetic energy stored in the magnetic field is

$$\frac{d}{dt}\left[\frac{1}{2}L(y(t))i^2(t)\right] = \frac{1}{2}i^2(t)\frac{dL(y(t))}{dt} + L(y(t))i(t)\frac{di(t)}{dt}$$

and the mechanical power output is $-f(t)\frac{dy(t)}{dt}$. The minus sign here is due to the opposite directions chosen for $f(t)$ and $y(t)$. Due to the conservation of energy, we have

$$i^2(t)\frac{dL(y(t))}{dt} + i(t)L(y(t))\frac{di(t)}{dt} = \frac{1}{2}i^2(t)\frac{dL(y(t))}{dt} + L(y(t))i(t)\frac{di(t)}{dt} - f(t)\frac{dy(t)}{dt}.$$

After canceling the equal terms in both sides, we obtain

$$f(t) = -\frac{1}{2}i^2(t)\frac{dL(y(t))}{dy} = \frac{rL_1}{2}\frac{i^2(t)}{y^2(t)}.$$

which relates the lifting force with the coil current and the ball position.